

BOOK REVIEW

Steven Mithen, *The language puzzle: Piecing together the six-million-year story of how words evolved*. New York: Basic Books, 2024. Pp. viii + 534.

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The Language Puzzle examines problems of linguistic theory related to the evolutionary origin of language knowledge. These problems are difficult because the evidence is remote and indirect. Nevertheless, what is notable in this contribution is that the overview of the research presents itself during a moment of opportunity, one of a realignment of hypotheses that cuts across the seemingly enduring and even widening divide in the language sciences. The proposals realign, between AND within Usage-based (U-B) and Universal Grammar (UG) approaches, navigating in an interesting way the proposed categories of continuity and discontinuity:

- (1) On the one hand, proposals recognize factors related to communicative ability and the participation of domain-general processes in the evolution of linguistic knowledge— continuity hypotheses.
- (2) In clear counter-position, discontinuity minimizes or denies the importance of these factors in accounting for the genesis of core grammatical competence: syntactic recursion, the Faculty of Language Narrow (FLN). The FLN thus has no relevant parallels in ancestral species to the knowledge of language in modern humans. There is no evidence for either homologs or analogs to FLN in other animals (Hauser et al. 2014).¹

Most interestingly, the role ascribed to communicative abilities and domain-general learning (pro and con) may apply to not only the debate on language evolution – in our species and among our ancestors – but also language development across just one lifetime. Here, a possible common ground (limited perhaps) on the role of communicative ability and domain-general learning in the emergence of language knowledge presents itself among proponents of both U-B and UG approaches. Thus, an opening for a new discussion between the opposing viewpoints is possible because it indicates or suggests a greater coherence, for both points of view.

[1] All sides can set aside, for now, the culturally acquired secondary attainments across history – the language abilities associated with the use of linguistic competence in the higher-order realms associated with literacy, the oral tradition, the poetic genres, etc. What Hauser et al. (2014) single out as the FLN would be the most circumscribed reckoning of ‘core grammar’. Other frameworks extend the boundary further and in different ways.

The book's proposal, following previous analysis of archaeological and linguistic findings and perhaps departing in some respects from the above suggestion, falls clearly within the continuity family of hypotheses. As we will see, this kinship relationship does not resolve the important internal disagreements within (1); but it greatly helps us to better understand the concepts across the board. Crucially, (1) and (2) also differ on the possible role that natural selection played in the emergence of language competence in modern humans.

Thus, the account of Continuity (U-B and UG) converges on a scenario of gradual evolution of language, guided by natural selection, in which there materialized a transition/adaptation between pre-linguistic communicative abilities and fully formed linguistic communicative abilities. Continuity (U-B) is well represented by *The Language Puzzle* and the hypothesis of Everett (2017), and Continuity (UG) by Pinker (2013). Discontinuity (2) favors a saltationist model of language emergence (not gradual evolution), is skeptical about natural selection, and discounts the importance of ancestral communicative abilities, as neither homologous nor analogous antecedents, as outlined above. While the discontinuity view is represented by only one proposal among researchers of language (Hauser et al. 2014), it represents a major and strongly dissident current on this question. See the review by Francis (2024) for a more complete summary of the different points of view.

For good reason, and helpful for the discussion, the chapters do not dwell on the topic of innateness. Steven Mithen saves time and spares us distraction. All models that accept a (biological) evolutionary schema (gradual or sudden, continuity/natural selection or discontinuity/saltation, U-B or UG) assume innate capacity at some level, the difference consisting of whether the relevant neural substrate is comprehensively domain-general or that the corresponding underlying structures are domain-specific to some degree. No known scientific hypothesis denies an evolved qualitative difference, of vast dimension, encoded in the genome, between modern humans and our primate cousins (surviving and extinct) regarding the domain-general capabilities. If core linguistic competence is entirely subserved by a domain-general faculty, as argued in the book, then language would be a compelling example. The great merit of Mithen's proposal, regardless of whether or not one agrees with it, is to pose this problem again, a much more difficult problem given that the emergence of domain-specific capabilities is much more easily accounted for, in theory, by processes of descent with modification.

In the first four chapters, the basic concepts are introduced and explained: The central idea when comparing features in evolution is that of distinguishing between the relationship of homology (in the book, termed 'ancestral' relationship) and analogy ('convergent evolution,' 69–70). The former considers a shared trait that can be traced to a common ancestor, the latter not, suggesting an independent emergence, an important finding nevertheless when it can be demonstrated. '[Independent] features' emerge as 'evolution finds the same solution to a common problem... long after the time of the last common ancestor' (70). Two lineages emerge that '[face] similar selective pressures for effective communication' (93),

for example; a similar, or analogous, feature appears independently in both. Birdsong and vocal communication in primates is perhaps a good example of analogy for our topic. Separate from the relationship of homology, the cases of convergent evolution are important to study. They allow us, for example, to focus attention on the factor of learning.

Darwin's (1981 [1871]) proposal of an ancestral musical protolanguage refers to a relationship of *HOMOLOGY* in regard to language. In contrast, birdsong is the closest *ANALOGY* to language in regard to the spontaneous process of acquisition or learning: an evolved instinct (cognitive faculty) for singing, with the species-specific songs, and their dialects, dependent on relevant input from parents (Fitch 2013). Vocal learning has evolved independently in three other clades of mammals and three clades of birds (494). Expressive and receptive capabilities plausibly confer survival advantage. Forward to the *Homo* genus, greater sensitivity on the part of infants to communicative vocalization of caregivers would have also been selected, an example of the co-evolution of general cognition and language.

More generally, again hypothetically, a homologous relationship could be shown to have appeared if evidence were to be presented, between modern humans and a lineage of archaic *Homo* with whom we shared a common ancestor. The proposed precursor, evolving from *H. heidelbergensis*, for example, may have possessed an evolving protolanguage, or even a single relevant feature, that both lineages inherited from this common ancestor. Or earlier, a pivotal protolanguage feature emerged in a common ancestor that a *H. heidelbergensis*-era species shared with another population, today also extinct. What the reader comes to appreciate from these chapters is that the current discoveries of ancient populations to which we can give a name are only a sample and reasonable approximation. Our closest living primate cousins evolved in a different direction, having diverged earlier. Later, in one of the lineages of our branch, an incipient capacity to acquire language emerged. *H. sapiens* would thus be the only surviving descendent among these closely related ancestors. Chapter 4 describes a different possible branching, but the basic idea should be the same.

Even though other traits that are being compared are not homologous, independently evolved communication systems of other animals are important to better understand. Convergent evolution provides evidence for how the forces of natural selection produce analogous results. All the while, within the framework of continuity hypotheses, we are mindful to distinguish between the two categories.

In an earlier assessment of the proposal for a holistic protolanguage (for example, by Wray 2002), Mithen (2006) reviewed counterproposals that suggest that the segmentation of holistic communication does not easily follow from the processing of internally undifferentiated utterances. But the recurring and expanding use of iconic word-like expressions, for example, of an onomatopoeic kind referencing vocalization of other animals and naturally occurring sounds, might have served as models for segmenting the parallel, predominantly holistic, system. The concluding Chapter 16 of the (2024) book outlines a scenario compatible with this possibility. Readers will take note that the channel for the evolving ancestral systems is the early

human voice and auditory mechanisms. Fitch (2013) proposes, similar to Mithen (2006), that the ancestral holistic system may have consisted of a ‘musical proto-language’ following a close reading of the original idea by Darwin in *The Descent*. In this model, incipient musical competencies and linguistic competencies formed part of an integrated vocal communicative system, that with time evolutionary forces would separate into independent faculties. The observation of shared features, in modern humans undergirded by partially overlapping neural circuits, could be taken as consistent with the hypothesis of Darwin and of Mithen (2006).

This review can conclude with a final reflection on the prospects of a productive exchange of views among proponents (from U-B and UG) of the continuity family of models, to then better understand the contrast with the discontinuity hypothesis. *The Language Puzzle* presents perhaps one of the strongest arguments in favor of the exclusive application of domain-general mechanisms across all aspects of language, evolutionary and developmental (6, 43, 192, 195, 216, 307, 365). Even so, the door is left open, in a number of the chapters, for considering the possibility that specialization and domain-specific processes should not be discounted, for example, as outlined in Mithen (2006).

In the (2024) discussion of aspects of both higher-order and automatic, System-1 type, processing, examples of modular or specialized knowledge structures are described. The hypothesis that linguistic competence and concepts maintain degrees of separation or mutual autonomy is presented as open to consideration (315–316, 328). Language, as is the case for other kinds of knowledge, is ‘distributed throughout the brain, which... has long-distance connections between areas of functional specialization... neural networks that develop under the combined influence of genes and environment’ (366). Chapter 11, in particular, focuses on competence structures that are domain-specific, evidenced for example by findings of double-dissociation (241). These specialized domains have been preserved in modern humans, with the progressive upgrading of greater interconnectivity. An account is needed then for why, specifically within the cognitive realm of linguistic competence, none of its component sub-domains came to be specialized in a similar way. On a related point, a follow-up discussion could expand on the relationship between the concepts of ‘poverty of stimulus’ and ‘learning bottleneck’ (188) and the interesting topic of learning biases in child cognitive development (204–212).

The emphasis here and in the other chapters, previewed in the subtitle of the book, is on word learning. A second edition could expand the summary of research to grammar. A plausible argument in this regard is that the former depends to a greater degree on domain-general declarative knowledge and cultural transmission than the latter.

Two pending research questions from the book suggest a common ground for dialogue:

- i. How, for example, do the networks of phonological representation and conceptual structure (meaning) interface? That is, through what kind of

intermediary network, or component domain, are interactions implemented? Everyone agrees: How ‘connections translate... remains unknown’ (235).

- ii. How should we better understand this important idea: a ‘long period of moulding by natural selection for speaking and hearing... must have gone hand in hand with the evolution of a linguistic capacity in the brain’ (95)?

There is broad agreement that the structure and functioning of the interfaces among the components of language knowledge and then with concept knowledge (i) are still not well understood. This indeed is a hard problem. We could then consider the research question about how natural selection formed the linguistic capacity, in all of its complexity (ii), in the spirit of a similar agreement: that this problem is still not well understood.

REFERENCES

- Darwin, C. 1981 (1871). *The descent of man, and selection in relation to sex*. Princeton, NJ: Princeton University Press.
- Everett, D. 2017. *How language began: The story of humanity's greatest invention*. New York: Norton.
- Fitch, W. T. 2013. Musical protolanguage: Darwin's theory of language evolution revisited. In J. J. Bolhuis & M. Everaert (eds.), *Birdsong, speech, and language: Exploring the evolution of mind and brain*. Cambridge, MA: MIT Press.
- Francis, N. 2024. The evolution of communication and language in the voices of nature. *Biology and Philosophy*, 39.11, 1–13.
- Hauser, M., C. Yang, R. C. Berwick, I. Tattersall, M. J. Ryan, J. Watumull, N. Chomsky & R. C. Lewontin. 2014. The mystery of language evolution. *Frontiers of Psychology*, 5, 401, 1–12.
- Mithen, S. 2006. *The singing Neanderthals: The origins of music, language, mind and body*. Cambridge: Harvard University Press.
- Pinker, S. 2013. Language as an adaptation to the cognitive niche. In M. H. Christiansen & S. Kirby (eds.), *Language evolution*, 16–37. Oxford: Oxford University Press.
- Wray, A. 2002. *The transition to language*. Oxford: Oxford University Press.

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